Symmetries in Physics — Exercise Sheet 5

Characters, tables, and projectors

Variant: Questions only

Exercise 1 Construct the character table of S_3 from first principles: find the classes, write down the two one-dimensional characters, and determine the third row using orthogonality alone. Verify column orthogonality afterwards.

Exercise 2 Compute the character tables of D_4 and of Q_8 and show they are *identical* (after matching classes). Conclude that the character table does not determine the group.

Exercise 3 S_4 permutes the coordinates of $\mathbb{C}^4$. Compute the character of this permutation representation, then use the character calculus to decompose it and to prove that its three-dimensional summand is irreducible.

Exercise 4 For the cyclic action of $\mathbb{Z}/3\mathbb{Z}$ on $\mathbb{C}^3$ (Sheet 3, Exercise 1), write down the three character projectors $P_k = \frac{1}{3}\sum_g \overline{\chi_k(g)}\,\Pi(g)$ and apply each to e_1; identify the images.

Exercise 5 (retrieval: parity through character projectors) Return to the parity action of $\mathbb{Z}/2\mathbb{Z}$ on $L^2(\mathbb{R})$ from Sheet 3, Exercise 5. Using the two characters and the character-projector formula from this lecture (proved for finite-dimensional spaces; for a group of order 2 the averaging makes sense on $L^2(\mathbb{R})$ directly, and you may use it in that form), compute both projectors and identify their images. Explain why each image is an isotypic component but is not itself an individual irreducible representation.

Exercise 6 The *tensor square* of a representation Π on V is the action $(\Pi\otimes\Pi)(g) = \Pi(g)\otimes\Pi(g)$ on $V \otimes V$; its *symmetric* and *antisymmetric squares* are the restrictions to the invariant subspaces $\mathrm{Sym}^2 V$ and $\Lambda^2 V$, spanned by the $e_ie_j + e_je_i$ $(i \le j)$, respectively the $e_ie_j - e_je_i$ $(i < j)$, writing e_ie_j for $e_i \otimes e_j$; both are invariant because $\Pi(g) \otimes \Pi(g)$ commutes with the swap $e_i \otimes e_j \mapsto e_j \otimes e_i$. (So if $\Pi(g)$ has eigenvalues $\{\lambda_i\}$, then $(\Pi \otimes \Pi)(g)$ has eigenvalues $\{\lambda_i\lambda_j\}_{i,j}$.)
(i) If Π has eigenvalue multiset $\{\lambda_i\}$ at g, show that these have characters

$$\chi_{\mathrm{Sym}^2}(g) = \tfrac{1}{2}\big(\chi(g)^2 + \chi(g^2)\big), \qquad \chi_{\Lambda^2}(g) = \tfrac{1}{2}\big(\chi(g)^2 - \chi(g^2)\big).$$

(ii) Apply both to the two-dimensional irreducible E of S_3 and decompose the results.

Exercise 7 Let Π be a d-dimensional representation of a finite group. Show $|\chi(g)| \le d$ for every g, with equality if and only if $\Pi(g)$ is a scalar multiple of the identity. Deduce that $\chi(g) = d$ exactly on the kernel of Π: the character detects faithfulness.